ITDS 124 Mathematics of Computing

# Chapter 1. Fundamentals

# 1.2 Operations on Sets

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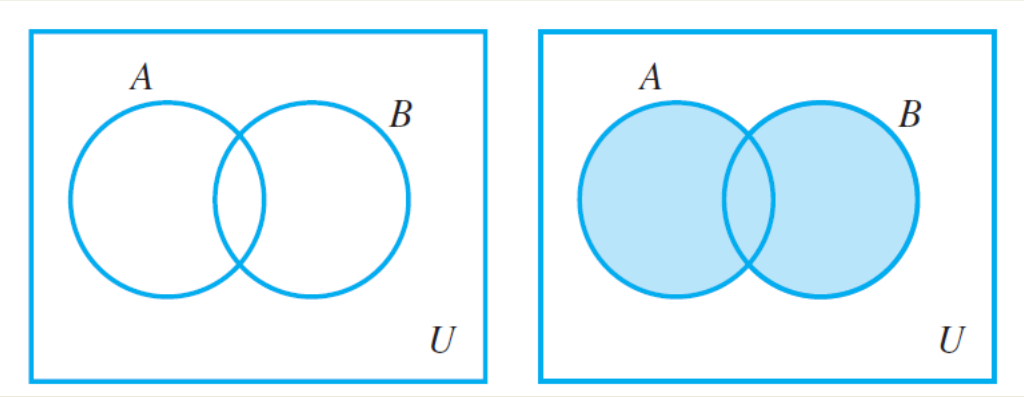
# Union

- If $A$ and $B$ are sets, we define their union as the set consisting of all elements that belong to $A$ or $B$ and denote it by $A \cup B$.
- $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

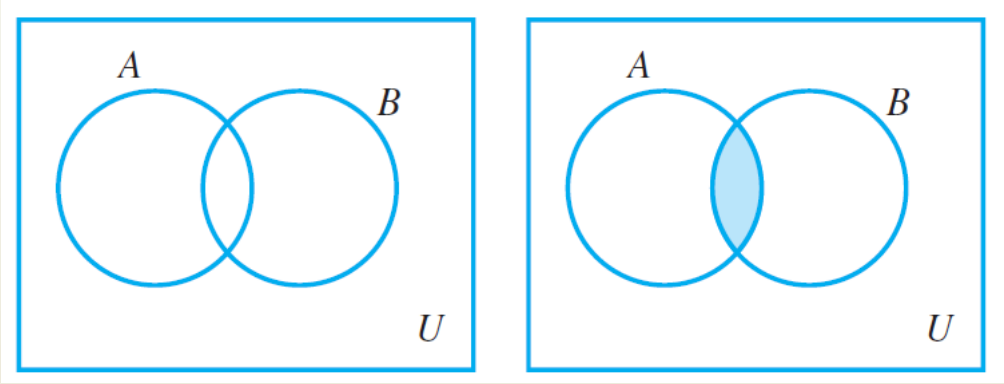



$A \cup B$

- Ex. Let $A = \{a, b, c, d, e, f\}$ and $B = \{b, d, r, s\}$.
  - $A \cup B$ is the set of all elements belong to either $A$ or $B$ = $\{a, b, c, d, e, f, r, s\}$

# Intersection

- If $A$ and $B$ are sets, we define their intersection as the set consisting of all elements that belong to both $A$ and $B$ and denote it by $A \cap B$.
- $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

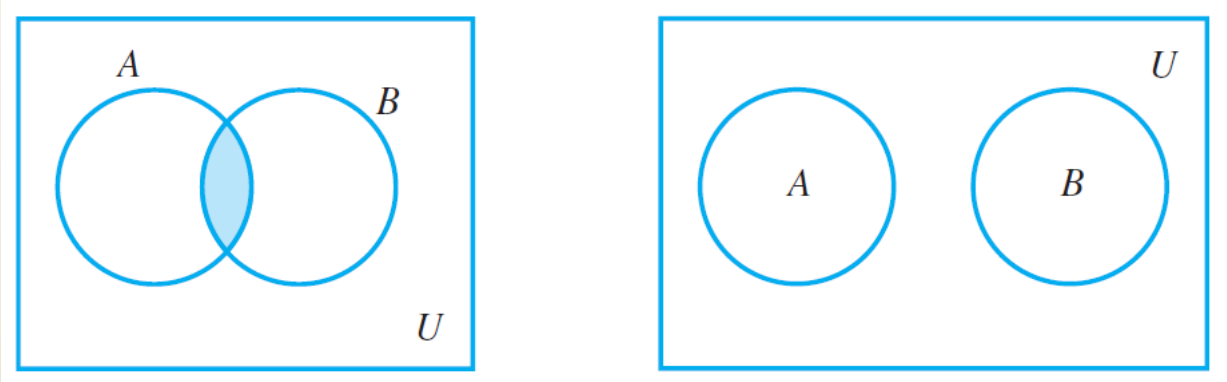



$A \cap B$

- Ex. Let $A = \{a, b, c, d, e, f\}$ $\quad B = \{b, d, r, s\}$ $\quad C = \{x, y, z\}$
  - $A \cap B$ is the set of all elements belong to both $A$ and $B$ = $\{b, d\}$
  - $A \cap C = \{\,\}$

# Disjoint sets

- Two sets are called disjoint sets if they have no common elements.
- Two sets A and B are disjoint if and only if their intersection $A \cap B = \emptyset$.

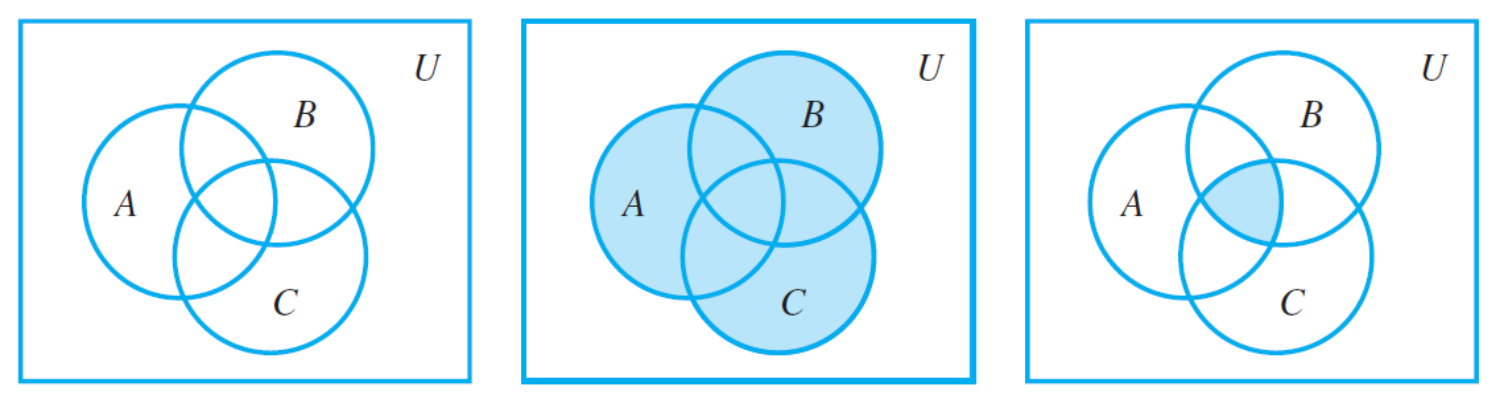



Not disjoint

Disjoint

# Union and Intersection for three or more sets

- $A \cup B \cup C = \{x \mid x \in A \text{ or } x \in B \text{ or } x \in C\}$
- $A \cap B \cap C = \{x \mid x \in A \text{ and } x \in B \text{ and } x \in C\}$

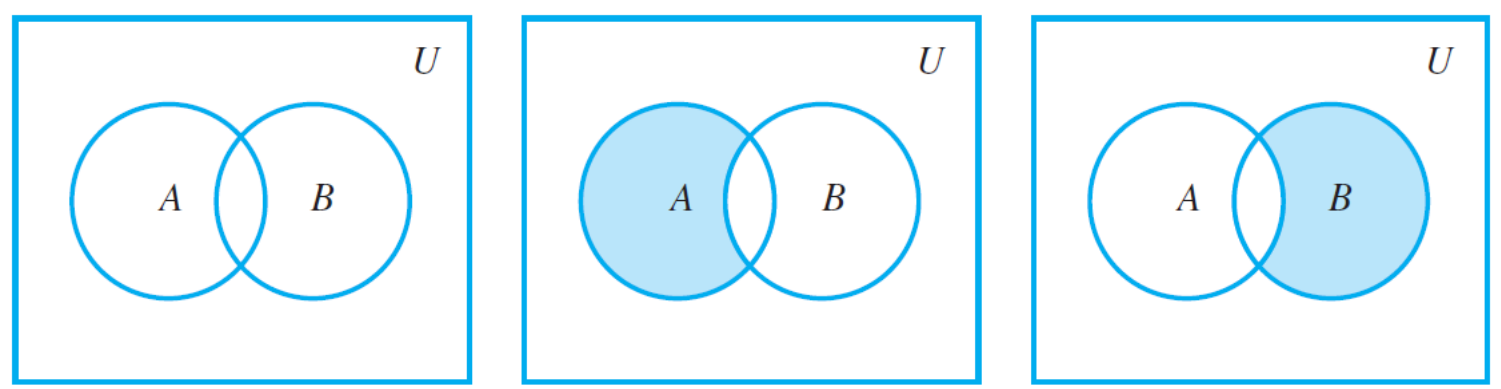



$A \cup B \cup C$ $\qquad$ $A \cap B \cap C$

- $A_1 \cup A_2 \cup A_3 \cup \cdots \cup A_n = \bigcup_{k=1}^{n} A_k$
- $A_1 \cap A_2 \cap A_3 \cap \cdots \cap A_n = \bigcap_{k=1}^{n} A_k$

# Complement

- If $A$ and $B$ are two sets, we define the complement of $B$ with respect to $A$ as the set of all elements that belong to $A$ but not to $B$.
- $A - B = \{x \mid x \in A \text{ and } x \notin B\}$




$A - B$ $\qquad$ $B - A$

- If $U$ is a universal set containing $A$, then $U - A =$ the complement of $A = \overline{A} = \{x \mid x \notin A\}$.

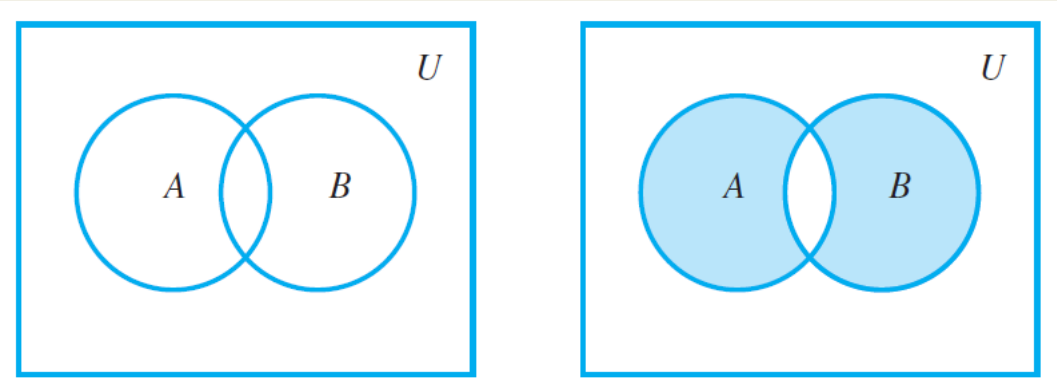
# Examples of complement

- Ex. Let $A = \{a, b, c\}$, $B = \{b, c, d, e\}$.
  - $A - B = \{a\}$
  - $B - A = \{d, e\}$

- Ex. Let $A = \{x \mid x \text{ is an integer and } x \leq 4\}$ and $U = \mathbb{Z}$.
  - $\overline{A} = \{x \mid x \text{ is an integer and } x > 4\}$

# Symmetric difference

- If $A$ and $B$ are two sets, we define their symmetric difference as the set of all elements that belong to $A$ or to $B$, but not to both $A$ and $B$.
- $A \oplus B = \{x \mid (x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A)\}$
- $A \oplus B = (A - B) \cup (B - A)$




$A \oplus B$

- Ex. Let $A = \{a, b, c, d\}$ and $B = \{a, c, e, f, g\}$
  - $A \oplus B = \{b, d, e, f, g\}$

# Algebraic properties of set operations

- Commutative properties
  - $A \cup B = B \cup A$
  - $A \cap B = B \cap A$
- Associative properties
  - $A \cup (B \cup C) = (A \cup B) \cup C$
  - $A \cap (B \cap C) = (A \cap B) \cap C$
- Distributive properties
  - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
  - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

# Algebraic properties of set operations

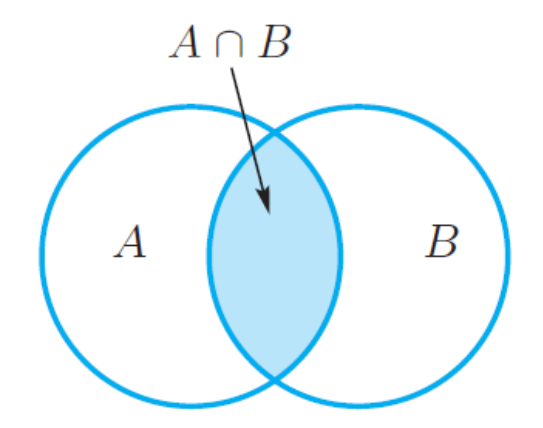
- Idempotent properties
  - $A \cup A = A$
  - $A \cap A = A$
- Properties of the complement
  - $\overline{\overline{A}} = A$
  - $A \cup \overline{A} = U$
  - $A \cap \overline{A} = \emptyset$
  - $\overline{\emptyset} = U$
  - $\overline{U} = \emptyset$
  - $\overline{A \cup B} = \overline{A} \cap \overline{B}$ ← De Morgan's law
  - $\overline{A \cap B} = \overline{A} \cup \overline{B}$ ← De Morgan's law

# Algebraic properties of set operations

- Properties of a universal set
  - $A \cup U = U$
  - $A \cap U = A$
- Properties of the empty set
  - $A \cup \emptyset = A$
  - $A \cap \emptyset = \emptyset$

# The addition principle

- Also called the inclusion-exclusion principle
- To compute $|A \cup B|$, we need to add $|A|$ and $|B|$, since $|A \cap B|$ is counted twice, so we need to subtract it one more.

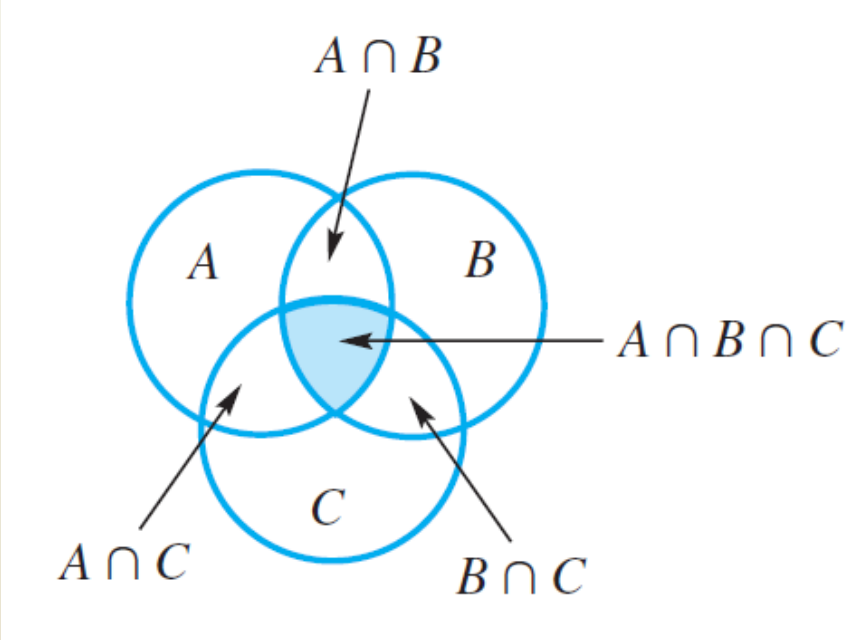



- $|A \cup B| = |A| + |B| - |A \cap B|$
- If $A$ and $B$ are disjoint sets, $A \cap B = \emptyset$, $|A \cap B| = 0$

$$|A \cup B| = |A| + |B|$$

# The three-set addition principle

- Let $A, B$ and $C$ be finite sets.
  - $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$



# Examples

- Ex1. Let $A = \{a, b, c, d, e\}$, $B = \{d, e, f, g, h\}$, $C = \{a, b, d, g, h, i\}$
  - Compute $|A \cup B|$
  - Compute $|A \cup B \cup C|$

# Examples

- Ex2. A computer company wants to hire 25 programmers to handle systems programming jobs and 40 programmers for applications programming. Of those hired, 10 will be expected to perform jobs of both types. How many programmers must be hired?

# Examples

- Ex3. A survey has been taken on methods of commuter travel. Each respondent was asked to check BUS, TRAIN, or AUTOMOBILE as a major method of traveling to work. More than one answer was permitted. The results reported were as follows: BUS 30 people, TRAIN 35 people, AUTOMOBILE 100 people, BUS & TRAIN 15 people, BUS & AUTOMOBILE 15 people, TRAIN & AUTOMOBILE 20 people, and all three methods 5 people. How many people completed a survey form?